

A Point-Like Prediction for Dark Matter at 12.8 GHz from a QCD-Anchored Vacuum Condensate: Theory and Haloscope Test Protocol

Oleg Kirichenko

Independent Researcher; urevich55@gmail.com
<https://github.com/infosave2007/vmf>

August 4, 2026

Abstract

The Null-Vector Gravity (NVG) framework, anchored to a single QCD input $M_{\Omega,0} = 859 \pm 8$ MeV fixed by lattice sigma terms, contains a pseudo-Goldstone mode θ of the vacuum condensate $\Phi = W e^{i\theta}$. If the same decay constant f_a that sets the θ -mode mass also generates Majorana neutrino masses through a QCD-anomaly seesaw, the neutrino sector fixes $f_a = 1.07 \times 10^{11}$ GeV with no further freedom. This single number then yields a *point* prediction for dark matter: a coherent pseudo-scalar of mass $m_\theta = 53.2$ μ eV, detectable as a microwave line at $f_\theta = 12.81$ GHz with photon coupling $g_{\theta\gamma\gamma} = 2.08 \times 10^{-14}$ GeV $^{-1}$ (KSVZ benchmark). The predicted product $m_\theta f_a$ falls on the QCD axion band within 2.6%, and the frequency lies in the currently unexplored gap between the published ORGAN mass windows (25–26 and 63–67 μ eV) — directly on the lower edge of the SMASH band (50–200 μ eV) that ORGAN has declared as its roadmap target. We present the derivation, a calibration of the haloscope signal formula against the published ORGAN signal estimate, and two concrete test protocols: (A) a resonant-cavity array and (B) a *fixed-position* dielectric booster stack, where the point-frequency prediction eliminates the scanning mechanism. For a coherent dark-matter fraction of 24% (misalignment only), protocol B reaches SNR = 5 in ~ 18 h at boost factor $\beta = 10^4$. Discovery and exclusion criteria are preregistered. All numerical values are reproducible from the accompanying scripts.

Keywords: axion dark matter, haloscope, dielectric haloscope, pseudo-Goldstone boson, vacuum condensate, neutrino mass, point prediction

Contents

1	Introduction	2
2	Theoretical Framework	2
2.1	The anchor: $M_{\Omega,0}$ from lattice sigma terms	2
2.2	The θ pseudo-Goldstone mode	2
2.3	The θ -seesaw: fixing f_a from the neutrino sector	3
2.4	Point prediction for the dark-matter mass	3
2.5	Coupling to photons	3
2.6	Relic abundance: the honest coherent fraction	4
2.7	Fifth force: the pseudoscalar null channel	4
3	Experimental Signature	4

4	Status of Published Searches	4
5	Test Protocol	5
5.1	Signal-power formula and its calibration	5
5.2	Protocol A: resonant cavity	5
5.3	Protocol B: fixed-position dielectric booster stack	6
5.4	Readout, statistics, and preregistered criteria	6
6	Honest Caveats	7
7	Conclusion	7

1 Introduction

Direct searches for QCD axion dark matter have so far produced exclusion limits in disjoint mass windows: ADMX covers $\sim 2.7\text{--}4.2\ \mu\text{eV}$ [13], HAYSTAC covers $\sim 17\text{--}20\ \mu\text{eV}$ [14], ORGAN has scanned $25.45\text{--}26.27\ \mu\text{eV}$ [17], $63.2\text{--}67.1\ \mu\text{eV}$ [15] and $107.42\text{--}111.93\ \mu\text{eV}$ [16], while MADMAX is under construction for the $40\text{--}400\ \mu\text{eV}$ range [20]. The intermediate region around $50\text{--}60\ \mu\text{eV}$ ($12\text{--}15\ \text{GHz}$) remains unexplored.

This note reports that the NVG/VMF framework [2], anchored to a single QCD input [1], yields a parameter-free prediction *exactly* in that gap: a dark-matter pseudo-scalar at $53.2\ \mu\text{eV}$. Because the prediction is a *point* in (m, g_γ) space rather than a band, the experimental problem simplifies drastically: no scanning, no tuning mechanism — a fixed-frequency integration.

Section 2 gives the derivation. Section 3 states the observable signature. Section 4 documents the gap in published searches. Section 5 defines the test protocol, including a calibration of the signal formula against published data and preregistered discovery/exclusion criteria. Section 6 lists the framework assumptions honestly.

2 Theoretical Framework

2.1 The anchor: $M_{\Omega,0}$ from lattice sigma terms

The framework is anchored to one dimensionful QCD input. Separating the nucleon mass into current-quark and nonperturbative components via the Feynman–Hellmann sigma terms ($\sigma_{\pi N} \approx 44\ \text{MeV}$, $\sigma_{sN} \approx 30\ \text{MeV}$), the vacuum value of the nonperturbative component is fixed at [1]

$$M_{\Omega,0} = 859 \pm 8\ \text{MeV}, \tag{1}$$

which sets the condensate energy density scale $\rho_c \approx 7.09 \times 10^4\ \text{MeV}/\text{fm}^3$ used throughout the construction. No cosmological or astrophysical input is used at any stage.

2.2 The θ pseudo-Goldstone mode

The NVG vacuum condensate is parametrized as $\Phi = W e^{i\theta}$, where the phase θ is a pseudo-Goldstone mode with the standard anomaly-generated potential. Its mass obeys the same relation as the QCD axion,

$$m_\theta = \frac{\sqrt{\chi_{\text{top}}}}{f_a}, \tag{2}$$

with the topological susceptibility $\chi_{\text{top}}^{1/4} = 75.5\ \text{MeV}$ from lattice QCD [9]. In the normalization $m_a(f_a = 10^{12}\ \text{GeV}) = 5.691\ \mu\text{eV}$ [9], this becomes $m_\theta = 5.691\ \mu\text{eV} \times (10^{12}\ \text{GeV}/f_a)$.

2.3 The θ -seesaw: fixing f_a from the neutrino sector

The framework hypothesis that makes the prediction parameter-free is the θ -seesaw: the same mode θ generates Majorana neutrino masses without right-handed neutrinos, through the two-loop QCD anomaly diagram coupling θ to the lepton current,

$$\mathcal{L}_{\theta\nu} = \left(\frac{\alpha_s}{4\pi}\right)^2 \frac{v_{\text{EW}}^2}{f_a} \nu_L^T C \nu_L + \text{h.c.} \quad (3)$$

This gives, for the heaviest neutrino,

$$m_3 = \left(\frac{\alpha_s}{4\pi}\right)^2 \frac{v_{\text{EW}}^2}{f_a}. \quad (4)$$

With $\alpha_s(M_Z) = 0.1184$, $v_{\text{EW}} = 246.22$ GeV and the oscillation-and-cosmology anchor $m_3 = 50.3$ meV (normal ordering; the implied sum $\sum m_\nu = 59$ meV passes the DESI DR2 bound $\sum m_\nu < 64$ meV unconditionally), Eq. (4) fixes

$$f_a = \left(\frac{\alpha_s}{4\pi}\right)^2 \frac{v_{\text{EW}}^2}{m_3} = 1.07 \times 10^{11} \text{ GeV} \quad (5)$$

with no remaining freedom. We stress that Eq. (3) is a *framework assumption* (the coefficient $(\alpha_s/4\pi)^2 \approx 8.9 \times 10^{-5}$ arises from the two-loop anomaly diagram); it is the single non-standard ingredient, and it is directly testable (Section 6).

2.4 Point prediction for the dark-matter mass

Inserting (5) into (2):

$$m_\theta = 5.691 \text{ } \mu\text{eV} \times \frac{10^{12}}{1.07 \times 10^{11}} = 53.2 \text{ } \mu\text{eV} \quad (6)$$

or, equivalently, a Compton frequency

$$f_\theta = \frac{m_\theta c^2}{h} = 12.81 \text{ GHz.} \quad (7)$$

Consistency check (no tuning). The two inputs above — m_θ from the lattice susceptibility and f_a from the neutrino sector — are independent. Their product must land on the QCD axion mass–coupling band $m_a f_a \simeq m_\pi f_\pi \sqrt{z}/(1+z) \approx 5.82 \times 10^{15} \text{ eV}^2$ [9]. The framework gives $m_\theta f_a = 5.67 \times 10^{15} \text{ eV}^2$: **agreement within 2.6%**, well inside the lattice uncertainty of the band. This agreement is a non-trivial consistency test of the construction, not an input.

2.5 Coupling to photons

The effective interaction is the standard axion-electrodynamics term [7]

$$\mathcal{L} \supset -\frac{g_{\theta\gamma\gamma}}{4} \theta F_{\mu\nu} \tilde{F}^{\mu\nu} = g_{\theta\gamma\gamma} \theta \mathbf{E} \cdot \mathbf{B}, \quad g_{\theta\gamma\gamma} = \frac{\alpha}{2\pi} \frac{|E/N - 1.92|}{f_a}. \quad (8)$$

For the KSVZ benchmark [5] ($E/N = 0$) and the DFSZ benchmark [6] ($E/N = 8/3$):

$$g_{\theta\gamma\gamma}^{\text{KSVZ}} = 2.08 \times 10^{-14} \text{ GeV}^{-1}, \quad g_{\theta\gamma\gamma}^{\text{DFSZ}} = 8.10 \times 10^{-15} \text{ GeV}^{-1}. \quad (9)$$

The model-dependent factor $|E/N - 1.92|$ is the only residual theory uncertainty on the coupling; the KSVZ value is used as the discovery benchmark.

2.6 Relic abundance: the honest coherent fraction

For a post-inflationary scenario with the natural initial misalignment $\theta_i = \pi/\sqrt{3}$, the standard misalignment estimate [8] gives

$$\Omega_\theta h^2 \simeq 0.12 \theta_i^2 \left(\frac{f_a}{10^{12} \text{ GeV}} \right)^{7/6} = 0.0291, \quad (10)$$

i.e. **24% of the local dark-matter density** ($\Omega_{\text{DM}} h^2 = 0.120$). The remaining density may reside in topological defects of the condensate sector; it does not contribute to the coherent haloscope signal. All signal powers below use the conservative coherent fraction $f_{\text{coh}} = 0.24$; the $f_{\text{coh}} = 1$ values are also quoted where relevant. A larger coherent fraction (e.g. from defect decay) would strengthen the signal proportionally.

2.7 Fifth force: the pseudoscalar null channel

The Compton range $\lambda_\theta = \hbar c/m_\theta = 3.72 \text{ mm}$ falls inside the sub-millimetre window of torsion-balance experiments [11]. Because θ is a *pseudoscalar*, the monopole–monopole Yukawa channel vanishes identically: unpolarized Eöt-Wash-class searches must see *nothing* at this range (null prediction, falsified by any monopole signal), while the spin-dipole channel with coupling m_N/f_a sits at the edge of current spin-polarized bounds. This constitutes an independent, low-cost cross-check of the sector.

3 Experimental Signature

The complete prediction is summarized in Table 1.

Table 1: Parameter-free prediction of the NVG θ -DM sector. No quantity in the right column is fitted to haloscope data.

Quantity	Predicted value	Origin
Decay constant f_a	$1.07 \times 10^{11} \text{ GeV}$	neutrino sector, Eq. (4)
Mass m_θ	$53.2 \text{ } \mu\text{eV}$	Eq. (2), lattice χ_{top}
Frequency f_θ	12.81 GHz	$m_\theta c^2/h$
Virial linewidth Q_a	$\sim 10^6$	halo velocity dispersion
Line width Δf_a	12.8 kHz	f_θ/Q_a
$g_{\theta\gamma\gamma}$ (KSVZ)	$2.08 \times 10^{-14} \text{ GeV}^{-1}$	Eq. (8)
$g_{\theta\gamma\gamma}$ (DFSZ)	$8.10 \times 10^{-15} \text{ GeV}^{-1}$	Eq. (8)
Coherent DM fraction	24%	misalignment, Sec. 2.6
$m_\theta f_a$ vs QCD band	−2.6% deviation	consistency check

The signal is a nearly monochromatic microwave excess at 12.81 GHz with $Q_a \sim 10^6$ intrinsic quality factor, linearly polarized along the applied magnetic field, scaling as B_0^2 , and tracking the sidereal modulation of the halo wind (a secondary discriminant).

4 Status of Published Searches

Table 2 compiles the published haloscope coverage around the predicted mass.

Two remarks. First, no published limit reaches $g = 2.1 \times 10^{-14} \text{ GeV}^{-1}$ anywhere near $53 \text{ } \mu\text{eV}$: the closest result (ORGAN at 25.5–26.3 μeV) is a factor ~ 13 weaker in coupling, at a different mass. Second, the ORGAN collaboration’s stated goal is to scan the SMASH band 50–200 μeV under a constant magnetic field [15]; the NVG point sits exactly at the lower edge of that band, making a targeted single-frequency run a natural addition to their program.

Table 2: Published haloscope mass windows around the NVG prediction. The predicted point (53.2 μeV) lies in the unexplored gap between the two ORGAN windows, at the lower edge of the SMASH band (50–200 μeV) [10] that ORGAN declares as its roadmap target [15].

Experiment	Mass window (μeV)	Frequency (GHz)	Best limit on $g_{a\gamma\gamma}$ (GeV^{-1})
ADMX [13]	2.66–3.31	0.64–0.80	KSVZ reached
HAYSTAC II [14]	~ 17.3 –19.5	4.18–4.71	$\sim 10^{-13}$
ORGAN [17]	25.45–26.27	6.15–6.35	2.8×10^{-13}
NVG prediction	53.2	12.81	2.08×10^{-14}
ORGAN [15]	63.2–67.1	15.3–16.2	3.0×10^{-12}
ORGAN 1b [16]	107.42–111.93	26.0–27.1	cogenesis ALP excluded
MADMAX [20, 23]	40–400 (design)	10–100	prototype phase

5 Test Protocol

5.1 Signal-power formula and its calibration

For a microwave cavity haloscope [7, 24],

$$P_{\text{sig}} = g_{\theta\gamma\gamma}^2 \frac{\rho_\theta}{m_\theta} B_0^2 V C \min(Q_L, Q_a), \quad (11)$$

where $\rho_\theta = f_{\text{coh}} \rho_{\text{DM}}$ with $\rho_{\text{DM}} = 0.4 \text{ GeV}/\text{cm}^3$, V is the cavity volume, C the mode form factor, and Q_L the loaded quality factor.

Calibration against published data. The ORGAN collaboration quotes, for its Phase 1a apparatus, $P_{\text{sig}} \simeq 2.3 \times 10^{-25} \text{ W}$ (KSVZ) and $3.1 \times 10^{-26} \text{ W}$ (DFSZ) for a 64 μeV axion [15]. Two checks:

1. *Apparatus-independent*: the coupling-coefficient ratio $(g_{\text{KSVZ}}/g_{\text{DFSZ}})^2 = (0.97/0.36)^2 = 7.26$ versus the published power ratio $2.3/0.31 = 7.42$ — agreement at the 2% level, confirming the normalization convention of Eq. (8).
2. *Absolute*: back-solving Eq. (11) from the published $2.3 \times 10^{-25} \text{ W}$ implies an apparatus product $VCQ \approx 0.44 \text{ m}^3$ (for $B_0 = 7 \text{ T}$), consistent with an ORGAN-scale cavity at 15.6 GHz with $Q \sim 10^4$ – 10^5 . The unit chain of the present calculation is thus validated against a real haloscope at the factor-of-two level.

For the dielectric booster concept [18, 19, 20] the analogous expression is

$$P_{\text{sig}} = g_{\theta\gamma\gamma}^2 \rho_\theta B_0^2 \frac{A\beta}{m_\theta^2}, \quad (12)$$

where A is the disk area and β the dielectric boost factor; both Eq. (11) and (12) have been re-derived within the Lorentz-reciprocity framework [24, 25].

5.2 Protocol A: resonant cavity

At $f_\theta = 12.81 \text{ GHz}$ the TM_{010} mode requires a cylinder of radius $R = x_{01}c/(2\pi f_\theta) = 8.95 \text{ mm}$. With length $L = 50 \text{ mm}$, $V = 12.6 \text{ cm}^3$, form factor $C = 0.69$, $B_0 = 10 \text{ T}$ and a superconducting (Nb) cavity with baseline $Q_L = 10^5$,

$$P_{\text{sig}} = 6.3 \times 10^{-25} \text{ W} \quad (f_{\text{coh}} = 0.24). \quad (13)$$

A single cavity reaches $\text{SNR} = 5$ only after ~ 48 years: it is a pathfinder, not the discovery configuration. The viable cavity route is a phase-locked array (Table 3); $Q_L = 10^6$ at 13 GHz is flagged optimistic (Nb BCS losses scale as ω^2), so the $Q = 10^5$ rows are the engineering baseline.

Table 3: Integration time to $\text{SNR} = 5$ for cavity arrays ($B_0 = 10$ T, $T_{\text{sys}} = 1$ K, $f_{\text{coh}} = 0.24$, per-cavity signal from Eq. (11)).

	$N = 64$	$N = 256$
$Q_L = 10^5$ (baseline)	276 days	69 days
$Q_L = 10^6$ (optimistic)	0.3 days	0.1 days

5.3 Protocol B: fixed-position dielectric booster stack

The dielectric haloscope [18] replaces the resonant cavity by ~ 80 dielectric disks (LaAlO_3 , $\varepsilon \approx 24$) in a strong magnetic field; constructive interference of the emitted photons provides the boost β . In a scanning experiment the disk positions must be moved mechanically — the technically hardest subsystem, now being commissioned [22]. **The point-frequency prediction eliminates this subsystem entirely:** at a known frequency the inter-disk spacing is fixed, and the stack becomes a static optical element.

With $D = 1.2$ m disks ($A = 1.13$ m², the low-mass-end MADMAX aperture), $B_0 = 10$ T and $f_{\text{coh}} = 0.24$, Eq. (12) gives (Table 4). The boost values $\beta = 10^4$ – 10^5 are the design goals of the MADMAX program [20]; the prototype results are at an earlier stage [23], which we state as the principal technical risk of this option.

Table 4: Fixed dielectric stack, $B_0 = 10$ T, $T_{\text{sys}} = 1$ K, readout bandwidth equal to the axion linewidth $\Delta f_a = 12.8$ kHz (the stack boost band $\gg \Delta f_a$).

Boost β	$P_{\text{sig}} (f_{\text{coh}} = 0.24)$	$P_{\text{sig}} (f_{\text{coh}} = 1)$	$t(\text{SNR} = 5)$
10^4	3.1×10^{-23} W	1.3×10^{-22} W	18 h
10^5	3.1×10^{-22} W	1.3×10^{-21} W	11 min

Protocol B is strictly simpler than the cavity array: no cryogenic tuning mechanism, no phase-locked multi-cavity readout, a single wide-bore magnet, horn antenna and a near-quantum-limited amplifier — while the required boost β is already the stated MADMAX design target in a frequency band that includes 12.81 GHz.

5.4 Readout, statistics, and preregistered criteria

Noise. $T_{\text{sys}} \approx 1$ K is adopted: millikelvin physical temperature plus a flux-driven Josephson parametric amplifier. This is not an assumption but demonstrated technology — ORGAN’s most recent phase operates exactly this way [17].

Statistical test. With the Dicke radiometer relation

$$\text{SNR} = \frac{P_{\text{sig}}}{k_B T_{\text{sys}}} \sqrt{\frac{t}{\Delta f}}, \quad (14)$$

the *preregistered* search window is $f_\theta \pm 0.5\%$ (covering any plausible framework uncertainty in m_3 and χ_{top}), resolved in bins of width Δf_a . Because the prediction is a point rather than a band, the look-elsewhere penalty is negligible; no trials factor beyond the preregistered window is applied.

Discovery criterion. A 5σ excess inside the preregistered window, persisting under independent re-taking, with the two corroborating checks: (i) power scaling $\propto B_0^2$ when the magnet current is varied; (ii) consistency of the line shape with the Maxwell–Boltzmann halo profile. Reproduction at a second apparatus with a different magnet is required before any claim.

Exclusion criterion. A null result at KSVZ-level sensitivity, $g_{\theta\gamma\gamma} \leq 2.1 \times 10^{-14} \text{ GeV}^{-1}$ across the preregistered window, excludes the θ -seesaw dark-matter hypothesis of Sec. 2.3 at $f_{\text{coh}} = 0.24$. Because exclusion curves assume the full local density, the effective bound on g for a coherent fraction f_{coh} is weaker by $1/\sqrt{f_{\text{coh}}} \approx 2.0$; this rescaling must be applied when interpreting any null. A null at DFSZ level ($8.1 \times 10^{-15} \text{ GeV}^{-1}$) would be the definitive test.

Side-channel cross-checks. The same f_a predicts $\sum m_\nu = 59 \text{ meV}$ (CMB+BAO, ongoing) and a $0\nu\beta\beta$ window $m_{\beta\beta} \in [0, 6.4] \text{ meV}$ (LEGEND-1000, nEXO): these provide framework-level corroboration independent of haloscopes.

6 Honest Caveats

1. **Framework assumption.** The θ -seesaw, Eq. (3), is a postulate of the NVG construction (motivated by the two-loop anomaly diagram), not a standard-model result. If Eq. (4) fails — e.g. m_3 from a future $0\nu\beta\beta$ /oscillation combination disagrees with 50.3 meV — the entire frequency prediction moves. The haloscope test is precisely the discriminator.
2. **Model dependence of the coupling.** The factor $|E/N - 1.92|$ in Eq. (8) is benchmarked to KSVZ/DFSZ; a different UV completion rescales $g_{\theta\gamma\gamma}$ without moving the frequency. The mass prediction is unaffected.
3. **Coherent fraction.** All powers use $f_{\text{coh}} = 0.24$ (misalignment only). If most of the sector’s density sits in incoherent defects, haloscope signals weaken accordingly; if defects decay into coherent modes, they strengthen. This is the dominant astrophysical uncertainty and is why the exclusion criterion carries the $1/\sqrt{f_{\text{coh}}}$ rescaling.
4. **Technology risk.** $Q_L = 10^6$ at 13 GHz and $\beta = 10^5$ are program goals, not demonstrated hardware. The baseline rows ($Q_L = 10^5$, $\beta = 10^4$) reflect the validated state of the art.

7 Conclusion

A single neutrino-sector input of the NVG/VMF framework fixes $f_a = 1.07 \times 10^{11} \text{ GeV}$, which through lattice-QCD physics yields a point dark-matter prediction: $m_\theta = 53.2 \text{ } \mu\text{eV}$, a microwave line at 12.81 GHz with KSVZ-level photon coupling. The point lands on the QCD axion band within 2.6% (consistency, not tuning), sits in the currently unexplored gap between published ORGAN windows, and is directly on that collaboration’s declared SMASH roadmap. Because the prediction is a point, the search needs no scanning: a fixed dielectric booster stack reaches discovery-level significance in under a day of integration at design boost. The protocol above is preregistered: discovery requires 5σ plus B^2 -scaling and cross-reproduction; exclusion at KSVZ sensitivity across the window closes the θ -seesaw dark-matter hypothesis. Either outcome is informative — which is the property a prediction must have to deserve an experiment.

Appendix A: Reproducibility

All numbers in this note are produced by scripts in the accompanying repository (<https://github.com/infosave2007/vmf>, Zenodo-archived releases):

- `generator_project/calc_theta_haloscope_design.py` — geometry, signal powers (Eqs. (11)–(12)), ORGAN calibration, integration times, published-search status;
- `generator_project/calc_theta_haloscope_v2.py` — frequency derivation, axion-band consistency check (2.6%);

- `verification/nvg_neutrino_seesaw.py` — Eq. (4), fixing f_a ; `verification/nvg_theta_fifth_force.py` — pseudoscalar null channel.

References

- [1] O. Kirichenko, *Lattice Sigma Terms as an Anchor for the Dense Nuclear Matter Equation of State*, Zenodo preprint, doi:10.5281/zenodo.20214457 (2025).
- [2] O. Kirichenko, *NVG-Research: Null-Vector Gravity and Vacuum Mass Fraction framework, verification suite and scripts*, <https://github.com/infosave2007/vmf>.
- [3] S. Weinberg, *A New Light Boson?*, Phys. Rev. Lett. **40**, 223 (1978).
- [4] F. Wilczek, *Problem of Strong P and T Invariance in the Presence of Instantons*, Phys. Rev. Lett. **40**, 279 (1978).
- [5] J. E. Kim, *Weak-Interaction Singlet and Strong CP Invariance*, Phys. Rev. Lett. **43**, 103 (1979); M. A. Shifman, A. I. Vainshtein, V. I. Zakharov, *Can Confinement Ensure Natural CP Invariance of Strong Interactions?*, Nucl. Phys. B **166**, 493 (1980).
- [6] A. R. Zhitnitsky, *On Possible Suppression of the Axion Hadron Interactions*, Sov. J. Nucl. Phys. **31**, 260 (1980); M. Dine, W. Fischler, M. Srednicki, *A Simple Solution to the Strong CP Problem with a Harmless Axion*, Phys. Lett. B **104**, 199 (1981).
- [7] P. Sikivie, *Experimental Tests of the Invisible Axion*, Phys. Rev. Lett. **51**, 1415 (1983).
- [8] J. Preskill, M. B. Wise, F. Wilczek, *Cosmology of the Invisible Axion*, Phys. Lett. B **120**, 127 (1983); L. F. Abbott, P. Sikivie, *A Cosmological Bound on the Invisible Axion*, Phys. Lett. B **120**, 133 (1983); M. Dine, W. Fischler, *The Not-So-Harmless Axion*, Phys. Lett. B **120**, 137 (1983).
- [9] G. Grilli di Cortona, E. Hardy, J. Pardo Vega, G. Villadoro, *The QCD axion, precisely*, JHEP **01** (2016) 034, arXiv:1511.02867.
- [10] G. Ballesteros, J. Redondo, A. Ringwald, C. Tamarit, *Standard Model-Axion-Seesaw-Higgs Portal Inflation. Five problems of particle physics and cosmology solved in one stroke*, arXiv:1610.01639 (2016).
- [11] I. G. Irastorza, J. Redondo, *New experimental approaches in the search for axion-like particles*, Prog. Part. Nucl. Phys. **102**, 89 (2018).
- [12] P. W. Graham, I. G. Irastorza, S. K. Lamoreaux, A. Lindner, K. A. van Bibber, *Experimental Searches for the Axion and Axion-Like Particles*, Annu. Rev. Nucl. Part. Sci. **65**, 485 (2015).
- [13] N. Du et al. (ADMX Collaboration), *A Search for Invisible Axion Dark Matter with the Axion Dark Matter Experiment*, Phys. Rev. Lett. **120**, 151301 (2018), arXiv:1804.05750.
- [14] X. Bai et al. (HAYSTAC Collaboration), *Dark Matter Axion Search with HAYSTAC Phase II*, Phys. Rev. Lett. **134**, 151006 (2025), arXiv:2409.08998.
- [15] A. Quiskamp et al. (ORGAN Collaboration), *Direct search for dark matter axions excluding ALPogenesis in the 63- to 67- μ eV range with the ORGAN experiment*, Sci. Adv. **8**, eabq3765 (2022).

- [16] A. Quiskamp et al. (ORGAN Collaboration), *Exclusion of Axionlike-Particle Cogenesis Dark Matter in a Mass Window above 100 μeV* , Phys. Rev. Lett. **132**, 031601 (2024), arXiv:2310.00904.
- [17] A. P. Quiskamp et al. (ORGAN Collaboration), *Near-quantum-limited axion dark matter search with the ORGAN experiment around 26 μeV* , Phys. Rev. D **111**, 095007 (2025), arXiv:2407.18586.
- [18] A. Caldwell et al. (MADMAX Working Group), *Dielectric Haloscopes: A New Way to Detect Axion Dark Matter*, Phys. Rev. Lett. **118**, 091801 (2017), arXiv:1611.05865.
- [19] A. J. Millar, G. G. Raffelt, J. Redondo, F. D. Steffen, *Dielectric Haloscopes to Search for Axion Dark Matter: Theoretical Foundations*, JCAP **01** (2017) 061, arXiv:1612.07057.
- [20] P. Brun et al. (MADMAX Collaboration), *A new experimental approach to probe QCD axion dark matter in the mass range above 40 μeV* , Eur. Phys. J. C **79**, 186 (2019), arXiv:1901.07401.
- [21] S. Knirck et al. (MADMAX Collaboration), *Simulating MADMAX in 3D: Requirements for Dielectric Axion Haloscopes*, JCAP **10** (2021) 034, arXiv:2104.06553.
- [22] B. Ary Dos Santos Garcia et al. (MADMAX Collaboration), *First mechanical realization of a tunable dielectric haloscope for the MADMAX axion search experiment*, JINST **19**, T11002 (2024), arXiv:2407.10716.
- [23] B. Ary Dos Santos Garcia et al. (MADMAX Collaboration), *First search for axion dark matter with a MADMAX prototype*, Phys. Rev. Lett. **135**, 041001 (2025), arXiv:2409.11777.
- [24] J. Egge, *Axion haloscope signal power from reciprocity*, JCAP **04** (2023) 064, arXiv:2211.11503.
- [25] J. Egge et al., *Experimental determination of axion signal power of dish antennas and dielectric haloscopes using the reciprocity approach*, JCAP **04** (2024) 005, arXiv:2311.13359.